- "Zhu, Wenhao, Hongyi Liu, Qingxiu Dong, Jingjing Xu, Shujian Huang, Lingpeng Kong, Jiajun Chen"
- "Lei Li"
  year: "2023"
  domain: "AI and Scholarship"
  subdomain: "Research Methods and Practices"
  tensions:
- "Technical Capability vs Organizational Capacity"
  concepts:
- "large language models"
- "generative AI"
  methodologies:
- "comparative"
- "empirical"
  stakeholders:
  []
  source: "arXiv preprint arXiv:2304"
  url: "https://arxiv.org/abs/2304.04675"
  status: "reviewed"
  type: "annotation"
  created: "2026-05-14"
  tags:
- "annotation"
- "ai-and-scholarship"
- "research-methods-and-practices"
- "method/comparative"
- "method/empirical"
- "topic/large-language-models"
- "topic/generative-ai"
- "tension/technical-capability-v-organizational-capacity"

Multilingual Machine Translation with Large Language Models: Empirical Results and Analysis

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Zhu, Wenhao, Hongyi Liu, Qingxiu Dong, Jingjing Xu, Shujian Huang, Lingpeng Kong, Jiajun Chen; Lei Li (2023)

arXiv preprint arXiv:2304

🔗 https://arxiv.org/abs/2304.04675 (https://arxiv.org/abs/2304.04675)

Annotation

Zhu et al. perform an empirical study that tests the multilingual translation performance of eight popular LLMs on 102 languages. As the authors explain, this is a particularly difficult task as it requires "semantic alignment between languages." While LLMs tend to perform surprisingly well at translation, "it is also unclear [sic] how LLM acquires translation ability and which factors affect LLM's translation ability". This study in multilingual machine translation (MMT) therefore seeks to answer two questions: how do LLMs perform MMT over massive languages and what factors affect their performance?

The results of the study suggest that GPT-4 generally outperforms its competitors but still falls short of Google Translate in some tests. Two key takeaways from this study is that "exemplars in the tail of the prompt have larger influence on an LLM's behaviour," meaning that the order in which exemplars are given within a prompt matter, and that cross-lingual translation pairs are particularly helpful exemplars to LLMs. The authors ultimately conclude that an "LLM can acquire translation ability in a resource-efficient way, which indicates [a] promising future of LLM in multilingual machine translation" as the technology evolves.

NB: As this research was conducted prior to Google's implementation of GenAI/LLMs into Google Translate in 2024, it sometimes uses Google Translate as a baseline/comparison point for LLM-powered translation that can be confusing without that context.

Connections

Domain: [[AI and Scholarship]] → Research Methods and Practices

Tensions: [[Technical Capability vs Organizational Capacity]]

Key Concepts: [[Large Language Models]] · [[Generative Ai]]

Methodologies: `comparative`, `empirical`

See Also

- [[Bail 6206 - Can-Generative-Artificial-Intelligence-Im]] — shared concepts: generative AI, large language models | shared tension: Technical Capability vs Organizational Capacity | same subdomain
- [[Aguiar 2024 - Final-Thoughts-Digital-Humanities-Looki]] — shared concepts: generative AI, large language models | shared tension: Technical Capability vs Organizational Capacity
- [[Mai 2024 - The-Use-of-ChatGPT-in-Teaching-and-Learn]] — shared concepts: generative AI, large language models | shared tension: Technical Capability vs Organizational Capacity
- [[Rapanta 2025 - Critical-GenAI-Literacy-Postdigital-Con]] — shared concepts: generative AI, large language models | shared tension: Technical Capability vs Organizational Capacity
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- [[Simon 2025 - Generative-AI-Quadruple-Deception-Tru]] — shared concepts: generative AI, large language models | shared tension: Technical Capability vs Organizational Capacity
- [[Xie 2025 - The-Social-Impact-of-Generative-LLM-Base]] — shared concepts: generative AI, large language models | shared tension: Technical Capability vs Organizational Capacity
- [[Yue 2025 - A-Practical-Guide-and-Assessment-on-Usin]] — shared concepts: generative AI, large language models | same subdomain

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